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$$\int \sinh^2 x dx$$

$$\text{Stel } \begin{cases} u = \sinh x \\ dv = \sinh x dx \end{cases} \Rightarrow \begin{cases} du = \cosh x dx \\ v = \cosh x \end{cases}$$

$$\text{Parti integratie: } \int u dv = uv - \int v du$$

$$= \sinh x \cosh x - \int \cosh x \cdot \cosh x dx$$

$$= \sinh x \cosh x - \int \cosh^2 x dx$$

$$\text{We weten dat } \cosh^2 x - \sinh^2 x = 1, \text{ dus } \cosh^2 x = 1 + \sinh^2 x$$

$$= \sinh x \cosh x - \int (1 + \sinh^2 x) dx$$

$$= \sinh x \cosh x - \int 1 dx - \int \sinh^2 x dx$$

$$= \sinh x \cosh x - x - \int \sinh^2 x dx$$

$$\int \sinh^2 x dx = \sinh x \cosh x - x - \int \sinh^2 x dx$$

$$\int \sinh^2 x dx + \int \sinh^2 x dx = \sinh x \cosh x - x$$

$$2 \int \sinh^2 x dx = \sinh x \cosh x - x + C$$

$$\int \sinh^2 x dx = \frac{\sinh x \cosh x - x}{2} + C$$

$$\int \sinh^2 x dx = \frac{\sinh x \cosh x}{2} - \frac{x}{2} + C$$

References